Replication Package Description for BPMN Assignment Design

Purpose

This replication package describes the instructional design and procedure used in the study to support reuse and adaptation in other BPM education contexts. It focuses on the structure of the assignment and verification mechanisms rather than on the specific process domain or task content.

Course Context

The study was conducted in an undergraduate Business Process Management course at a technical university. Students had previously been introduced to core BPMN elements, including activities, sequence flows, message flows, gateways, events, and lanes. The assignment was implemented within a regular course setting and integrated into the standard curriculum.

Assignment Structure

Students completed an iterative BPMN modelling assignment consisting of three versions:

- V1 – an initial BPMN model created individually after the first instructional session,
- V2 – a revised model produced after standardized minimal feedback and structured self-verification,
- V2-AI – an optional AI-assisted alternative version created by a volunteer subgroup.

The V2-AI submission did not replace the revised V2 model and was treated as an exploratory learning activity rather than a performance-driven or mandatory task.

Instructional Scaffolds

All students used the same verification scaffolds when revising their models:

- a rubric-aligned BPMN verification checklist,
- three mandatory scenario-based test cases (happy path, rejection/error scenario, exception or boundary case),
- a change log documenting key revisions between V1 and V2.

Students submitting the optional V2-AI version additionally completed an AI Usage Statement documenting their goals, verification steps, and examples of accepted and rejected AI suggestions.

Scope of Replication

The replication package supports reuse of the instructional design, verification scaffolds, and evaluation approach. It does not include the original process descriptions, student submissions, or domain-specific task content. Educators are expected to adapt the process scenario to their own teaching context while retaining the iterative structure and verification-centred learning design. Detailed task descriptions can be adapted freely by educators to their local context and are therefore not included as part of the replication package.

Appendix 1. - the BPMN verification checklist used by students during iterative model revision and scenario-based test cases for BPMN model validation

Appendix 2. - the change logs template documenting model revisions between assignment iterations

Appendix 3. - the template of the AI Usage Statement completed for AI-assisted submissions

Appendix 4. - the BPMN error taxonomy applied for fine-grained error analysis

Appendix 5. - the rubric used for evaluating BPMN model quality

Appendix 1. BPMN Verification Checklist (+**Scenario-Based Test Cases for BPMN Model Validation**)

Complete the checklist for V2 (and if you're working on V2 AI, complete the one for V2 AI as well). Tick YES/NO and add a brief note where it says "Comment", e.g.: the diagram has not been corrected; there are still errors in it.

A. Diagram: participants, roles, steps, clarity	**YES**	**NO**
I have identified all participants in the process and their business roles (no 'anonymous' actors).		
The process steps are complete in terms of the description (no key activities are missing).		
The diagram is clear: no backtracking, no 'wavy' lines, no spaghetti effect.		
All elements are labelled; (optionally) element numbering is consistent..		
Comment:		
B. Message flow and collaboration between private diagrams		
For every transfer between participants, there is a correct message flow (not a sequence flow).		
There are no message flows "going nowhere" (source and destination are unambiguous).		
Collaboration between private diagrams is consistent: missing flows have been completed in accordance with the process logic.		
Each participant has a meaningful "scope" (I do not model everything in a single pool/lane if there are separate actors).		
Comment:		
C. Events: start, end, intermediate (including boundary)		
Start		
The start event has no incoming flow.		
The start event type is correct (not "default" if the process is initiated by a message/time/condition).		
Intermediate + boundary		
Intermediate events are selected correctly (the type makes sense in the scenario).		
Boundary events: I have correctly indicated whether they are interrupting or non-interrupting.		
End		
End events are selected correctly (e.g. message end, error end, etc., where appropriate).		
I use terminate end exclusively for premature termination, not for the happy flow.		
Comment:		
D. Gates: split/merge, data/event-based		
Split gates are selected correctly: XOR/AND/event-based in accordance with the logic.		
Merge gates are correct and logically close the branches.		
Data-driven vs event-driven gates are used correctly (I do not confuse event-based with XOR).		
Transition conditions are unambiguous (if I use conditions).		
Comment:		

E. Logic mini-tests (3 test cases — mandatory): for each test, write one sentence: “which way does the process go and why”.
Test 1 (happy flow):
Test 2 (rejection / error):
Test 3 (exception / timeout / no data):

Appendix 2. Change Logs Template for Iterative BPMN Modelling

Mini Change Log (V1)

What is the biggest uncertainty in V1? ___

Where do you expect bugs to occur? (multiple choice)

	message flows
	events
	gates
	readability
	other:

Change Log (V2)

1. Top 3 changes to the model compared to V1:
   1) ___
   2) ___
   3) ___
2. Which area was the most difficult? (one answer):

	Diagram/readability/roles
	Message flows/collaboration
	Events (start/end/intermediate/boundary)
	Gates (split/merge, data/event-based)

3. What contributed most to the improvement in V2? (one answer):

	checklist
	test cases
	own analysis
	instructor's feedback (if any)
	other:

4. One risk/error you are still unsure about: ___

Appendix 3. AI Usage Statement Template

"best possible model with AI"

1. Tool: ___
2. Frequency of use (1/occasionally–5/intensively): ___
3. What did I use the AI for? (max 3):

	gateways
	events (including boundary events)
	message flows/collaboration
	test cases / logic validation
	completeness of steps and roles
	other:

4. One AI suggestion I rejected and why (1 sentence): ___
5. One AI suggestion I implemented, and how I verified it (1 sentence): ___

Appendix 4. BPMN Error Taxonomy Used for Model Analysis

Purpose

This error taxonomy was used for fine-grained coding of student BPMN models and for analysing error-pattern changes across assignment iterations (V1, V2, and V2-AI). The taxonomy consists of ten error codes (E1–E10), applied consistently across all model versions.

Table 1. Error Codes and Definitions

Error ID	Label	Description
E1	identification of participants and roles	Errors related to incorrect or incomplete identification of process participants, pools, or lanes
E2	inclusion of all steps in the process, consistency with the description	Omission of required activities or inconsistencies between the BPMN model and the textual process description.
E3	readability (no waviness, no spaghetti effect)	Layout and readability issues that hinder process comprehension.
E4	labels and steps prior to sending messages and making decisions	Missing or unclear activity labels preceding message exchanges or decision points.
E5	correct use of start and end events	Incorrect, missing, or semantically inappropriate start or end events.
E6	exceptions vs decisions	Confusion between exception handling and decision logic.
E7	correct split/merge gates	Incorrect use or pairing of gateway splits and merges.
E8	inclusion of all message flows	Missing message flows required for inter-participant communication.
E9	correct modelling of message send/receive events	Incorrect use or placement of send/receive message events.
E10	completion of process flows (sequential)	Incomplete sequential flow from start to proper termination.

Use in Analysis

Each BPMN model (V1, V2, V2-AI) was coded for the presence of the above error codes. Error intensities were aggregated per code and compared across assignment iterations to identify learning effects and AI-related changes in modelling behaviour.

Error-Code Aggregation

Ten error codes (E1–E10) were applied to each model). For statistical analysis, error codes were aggregated into conceptually related groups aligned with rubric dimensions:

E1–E2: participants, roles, and process completeness,
E3–E4: readability and labeling,
E5–E6: event correctness and exception handling,
E7: gateway split/merge logic,
E8–E9: message flows and communication semantics,
E10: sequential completion of process flows.

Error intensities were computed per model and aggregated across students for each version.

Normality Assessment

For each aggregated error-code group and each model version (V1, V2, V2-AI), compliance with normal distribution assumptions was assessed using the Shapiro–Wilk test.

In most cases, the null hypothesis of normality was rejected, indicating non-normal distributions of scores and error intensities.

Statistical Tests

Given the ordinal nature of rubric scores, unequal group sizes, and non-normal distributions, non-parametric methods were applied throughout:

- Wilcoxon signed-rank test for paired comparisons:
  - V1 vs. V2 (full cohort),
  - V2 vs. V2-AI (volunteer subgroup).
- Effect sizes (r) were calculated for paired comparisons to estimate the magnitude of observed effects.

The significance level for all tests was set at $\alpha = 0.05$.

Interpretation Strategy

Statistical results were interpreted descriptively and comparatively, consistent with the design-based and exploratory character of the study. Emphasis was placed on:

- direction and consistency of changes across iterations,
- relative improvements across modelling dimensions,
- alignment between rubric-based improvements and reductions in specific error codes.

Statistical significance was used to support observed learning trends rather than to test narrowly defined causal hypotheses.

Software Environment

All statistical analyses were conducted using R (version 4.6.0). Visualisations included slope charts, paired dot plots, and heatmaps to support interpretation of individual-level and dimension-level changes

Appendix 5. Rubric for BPMN Model Evaluation

Purpose

This rubric was used to evaluate BPMN model quality for all submitted versions (V1, V2, V2-AI).

Rubric Structure

Dimension 1: Diagram Structure and Readability (0–2 points)

0 – major structural issues, unclear or incomplete flow

1 – mostly correct structure with minor issues

2 – clear, complete, and readable process structure

Dimension 2: Message Flows and Collaboration (0–1.5 points)

0 – incorrect or missing collaboration semantics

0.75 – partially correct use of message flows

1.5 – correct and consistent collaboration modelling

Dimension 3: Events (0–1 point)

0 – incorrect or missing events

0.5 – partially correct event usage

1 – correct and semantically appropriate events

Dimension 4: Gateways (0–0.5 points)

0 – incorrect or missing gateway logic

0.25 – mostly correct logic with minor issues

0.5 – correct and consistent gateway usage

Maximum score: 5 points. The same rubric was applied consistently across all model versions.